

Contingency Table (3) (분할표 (3))

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Today's Topics

1. Cochran-Mantel-Haenszel

- CMH Statistic
- Breslow-Day Statistic

2. R code for Contingency Table

3. Generalized Linear Model (Fundmental Structure)

- Components
- GLM for Binary Data

Cochran-Mantel-Haenszel

Test of conditional independence and test of homogeneous association for the K conditional odds ratios in $2 \times 2 \times K$ table

A method of combining the sample odds ratios obtained from K partial tables to obtain a single summarized measure of partial association

Cochran-Mantel-Haenszel (Cont'd)

(Example) Survival of rabbits infected with β -hemolytic streptococcus when penicillin is administered immediately versus after 1 hour and 30 minutes

페니실린	시간	생존여부		합
		치료	죽음	
1/8	즉시	0	6	6
	이후	0	5	5
1/4	즉시	3	3	6
	이후	0	6	6
1/2	즉시	6	0	6
	이후	2	4	6
1	즉시	5	1	6
	이후	6	0	6
4	즉시	2	0	2
	이후	5	0	5

Cochran-Mantel-Haenszel (Cont'd)

Null hypothesis

When Z is given, X and Y are conditionally independent = for all k , $\theta_{XY(K)} = 1$

Sampling model

Subtables are independent and each subtable follows a multinomial distribution with fixed sample size

In each subtable, given fixed row totals $\{n_{1+k}, n_{2+k}\}$ and column totals $\{n_{+1k}, n_{+2k}\}$, n_{11k} follows the hypergeometric distribution

Cochran-Mantel-Haenszel (Cont'd)

Under the null hypothesis

$$\text{Mean: } \mu_{11k} = E(n_{11k}) = \frac{n_{1+k}n_{+1k}}{n_{++k}} \quad E(X) = \frac{nt}{n+m}, \text{var}(X) = \frac{nmt(n+m-t)}{(n+m)^2(n+m-1)}$$

$$\text{Variance: } V(n_{11k}) = \frac{n_{1+k}n_{2+k}n_{+1k}n_{+2k}}{n_{++k}^2(n_{++k}-1)}$$

In the k -th subtable, if $\theta_{XY(K)} > 1$, then $n_{11k} - \mu_{11k} > 0$ is expected

If the odds ratio in each subtable is greater than 1, the sum of such differences tends to be a relatively large positive value

If it is less than 1, the sum tends to be a negative value

Cochran-Mantel-Haenszel (Cont'd)

CMH test statistic

$$CMH = \frac{\left\{ \sum_k (n_{11k} - \mu_{11k}) \right\}^2}{\sum_k V(n_{11k})} \quad \text{or} \quad \frac{\left\{ \sum_k (n_{11k} - \mu_{11k}) - \frac{1}{2} \right\}^2}{\sum_k V(n_{11k})} \sim \chi_1^2$$

When $n_{11k} - \mu_{11k}$ are all positive or all negative, the statistic tends to be larger than when the values are mixed

When the heterogeneity of association among subtables is large, the CMH test is inadequate

When the association across subtables is homogeneous, the CMH test is superior to conducting individual tests on each subtable

If $\sum_{k=1}^K n_{11k}$ is sufficiently large, the distribution of n_{11k} approximates the normal distribution under the null hypothesis, regardless of the size of each n_{11k}

Cochran-Mantel-Haenszel (Cont'd)

(Example) Penicillin and Rabbits

페니실린	1/8	1/4	1/2	1	4
μ_{11k}	0	1.5	4.0	5.5	2
$V(n_{11k})$	0	27/44	32/44	11/44	0

$$\sum n_{11k} = 16, \sum \mu_{11k} = 13, \sum V(n_{11k}) = 70/44$$

$$CMH = 3.93 \text{ (0.047)}$$

Cochran-Mantel-Haenszel (Cont'd)

(Example) Smoking and Lung Cancer by Chinese Cities

Center	Treatment	Lung Cancer		OR	μ_{11k}	$\text{Var}(\mu_{11k})$
		Yes	No			
Beijing	Smoking	126	100	2.20	113.0	16.9
	Non-smoking	35	61			
Shanghai	Smoking	908	688	2.14	773.2	179.3
	Non-smoking	497	807			
Shenyang	Smoking	913	747	2.18	799.3	149.3
	Non-smoking	336	598			
Nanjing	Smoking	235	172	2.85	203.5	31.1
	Non-smoking	58	121			
Harbin	Smoking	402	308	2.32	355.0	57.1
	Non-smoking	121	215			
Zhengzhou	Smoking	182	156	1.59	169.0	28.3
	Non-smoking	72	98			
Taiyea	Smoking	60	99	2.37	53.0	9.0
	Non-smoking	11	43			
Nanchang	Smoking	104	89	2.00	96.5	11.0
	Non-smoking	21	36			

$$CMH = 280.1 (0.000)$$

Rejection of the null hypothesis of conditional independence
= Smoking and lung cancer are closely related across regions

Cochran-Mantel-Haenszel (Cont'd)

Estimation of Common Odds Ratio

Estimating the strength of association

When associations across all subtables have relatively stable values, estimate the common odds ratio

Mantel-Haenszel (MH) Estimator of Common Odds Ratio

In $2 \times 2 \times K$ contingency table, assuming $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$

$$\hat{\theta}_{MH} = \frac{\sum_k (n_{11k}n_{22k}/n_{++k})}{\sum_k (n_{12k}n_{21k}/n_{++k})}$$

$$\log(\hat{\theta}_{MH}) \sim N(\log(\theta_{XY}), \sigma^2(\log(\theta_{XY})))$$

Cochran-Mantel-Haenszel (Cont'd)

Estimated Variance of $\log(\hat{\theta}_{MH})$

$$\begin{aligned}\hat{\sigma}^2[\log(\hat{\theta}_{MH})] = & \frac{\sum (n_{11k} + n_{22k})(n_{11k}n_{22k})/n_{++k}^2}{2\left(\sum n_{11k}n_{22k}/n_{++k}\right)^2} \\ & + \frac{\sum [(n_{11k} + n_{22k})(n_{12k}n_{21k}) + (n_{12k} + n_{21k})(n_{11k}n_{22k})]/n_{++k}^2}{2\left(\sum n_{11k}n_{22k}/n_{++k}\right)\left(\sum n_{12k}n_{21k}/n_{++k}\right)} \\ & + \frac{\sum (n_{12k} + n_{21k})(n_{12k}n_{21k})/n_{++k}^2}{2\left(\sum n_{12k}n_{21k}/n_{++k}\right)^2}\end{aligned}$$

Cochran-Mantel-Haenszel (Cont'd)

(Example) Penicillin and Rabbits

페니실린	1/8	1/4	1/2	1	4
μ_{11k}	0	1.5	4.0	5.5	2
$V(n_{11k})$	0	27/44	32/44	11/44	0

$$\hat{\theta}_{MH} = 7.0, \log(\hat{\theta}_{MH}) = 1.95, \hat{\sigma}(\log(\hat{\theta}_{MH})) = 0.98$$

95% confidence interval for θ_{XY} :

$$[e^{1.95 - 1.96 \times 0.98}, e^{1.95 + 1.96 \times 0.98}] = [1.03, 47.85]$$

Cochran-Mantel-Haenszel (Cont'd)

(Example) Smoking and Lung Cancer in Chinese Cities

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	Non-smoking	11	43			
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	Non-smoking	21	36			

$$\hat{\theta}_{MH} = 2.17, 95\% \text{ confidence interval} = [1.98, 2.38]$$

The odds of smokers developing lung cancer are about twice the odds for non-smokers

Breslow-Day Statistic

$$\sum_{k=1}^K \frac{(a_k - E(a_k; \hat{\theta}_{MH}))^2}{\text{var}(a_k; \hat{\theta}_{MH})} \sim \chi^2(df = K - 1)$$

Null hypothesis: Homogeneous association, i.e., for all k , $\theta_{XY(K)} = \theta$

Computation of $E(x_i; \hat{\theta}_{MH})$, $\text{var}(x_i; \hat{\theta}_{MH})$

$$\frac{E_k(n_{2k} - m_{1k} + E_k)}{(n_{1k} - E_k)(m_{1k} - E_k)} = \hat{\theta}_{MH}$$

$$V_k = \frac{1}{\left(\frac{1}{E_k} + \frac{1}{n_{1k} - E_k} + \frac{1}{m_{1k} - E_k} + \frac{1}{n_{2k} - m_{1k} + E_k} \right)}$$

Tarone's adjustment

$$\sum_{k=1}^K \frac{(x_k - E(x_k; \hat{\theta}_{MH}))^2}{\text{var}(x_k; \hat{\theta}_{MH})} - \left\{ \sum_{k=1}^K (x_k - E(x_k; \hat{\theta}_{MH})) \right\}^2 \left\{ \sum_{k=1}^K \text{var}(x_k; \hat{\theta}_{MH}) \right\}^{-1} \sim \chi^2(df = K - 1)$$

The Breslow-Day statistic, compared to the MLE, uses the relatively inefficient $\hat{\theta}_{MH}$

Taylor expansion is used to adjust the efficiency (correcting for small test power)

Breslow-Day 통계량 (Cont'd)

Remark

- CMH test

① H_0 accepted $\Rightarrow$ Conditional independence, $\theta = 1$

② H_0 rejected $\Rightarrow$ Perform Breslow-Day test

- Breslow-Day test

① H_0 accepted $\Rightarrow$ Homogenous association; estimate association using $\hat{\theta}_{MH}$

② H_0 rejected $\Rightarrow$ Interaction exists between X 와 Z ; estimate $\hat{\theta}_{XY(1)}, \dots, \hat{\theta}_{XY(K)}$ separately

R Code

Odds Ratio

약	심근경색		합
	유	무	
위약	189	10,845	11,034
아스피린	104	10,933	11,037

```
### Table 1 (2x2)
df1 <- matrix(c(104, 10933, # Aspirin: Event, NoEvent
               189, 10845), # Placebo: Event, NoEvent
              nrow = 2, byrow = TRUE,
              dimnames = list(Treatment = c("Aspirin", "Placebo"),
                              Outcome   = c("Event", "NoEvent")))
```

df1

```
# Inference for Odds Ratio
a <- df1[1,1]; b <- df1[1,2]; c <- df1[2,1]; d <- df1[2,2]
OR_hat <- (a*d)/(b*c); OR_hat
se_logOR <- sqrt(1/a + 1/b + 1/c + 1/d); se_logOR
alpha <- 0.05; z <- qnorm(1 - alpha/2)
CI_OR <- exp(c(log(OR_hat) - z*se_logOR, log(OR_hat) +
               z*se_logOR)); CI_OR
c(OR = OR_hat, L = CI_OR[1], U = CI_OR[2])
```


R Code (Cont'd)

Independence Test

약	심근경색		합
	유	무	
위약	189	10,845	11,034
아스피린	104	10,933	11,037

```
# Residual
obj <- chisq.test(df1, correct = FALSE)
obj$expected      # Expected
obj$residuals     # Pearson residuals: (O-E)/sqrt(E)
obj$stdres        # Adjusted (standardized) residuals ~ N(0,1)
```

```
# Fisher's Exact Test
fisher.test(df1, alternative="two.sided", conf.int=T)
```

```
# Pearson chi-square ( $X^2$ ) — when sample size is large
chisq.test(df1, correct = FALSE)
```

```
# Yates continuity correction - when sample size is small
chisq.test(df1, correct = TRUE)
```

```
# Score test
prop.test(x = c(a, c), n = c(a+b, c+d), correct = FALSE)
```

```
# LRT
E <- outer(rowSums(df1), colSums(df1)) / sum(df1)
G2 <- 2 * sum(df1 * log(df1 / E)); G2
df <- (nrow(df1) - 1) * (ncol(df1) - 1); df
p_G2 <- pchisq(G2, df, lower=F); p_G2
c(G2 = G2, df = df, p.value = p_G2)
```

R Code (Cont'd)

2x3 Table

성별	소속정당		
	민주당	공화당	무소속
여성	279	225	73
남성	165	191	47
합	444	416	120

```
### Table 2 (2x3)
df2 <- matrix(c(279, 225, 73, # Female: Democrats, Republicans, Independents
                165, 191, 47), # Male: Democrats, Republicans, Independents
              nrow = 2, byrow = TRUE,
              dimnames = list(Gender = c("Male", "Female"),
                              Party = c("Democrats", "Republicans", "Independents")))

df2

# Pearson chi-square (X^2) — when sample size is large
chisq.test(df2, correct = FALSE)
```

R Code (Cont'd)

Ordinal Data

알코올량	기형없음	기형있음	합계	기형비율(%)	수정잔차
0	17066	48	17114	0.28	-0.18
<1	14464	38	14502	0.26	-0.71
1-2	788	5	793	0.63	1.84
3-5	126	1	127	0.79	1.06
5<	37	1	38	2.63	2.71

```
# Pearson product-moment correlation with freq weights
require(weights)
r <- wtd.cors(x=rep(dat$AlcoholUsage, dat$Freq),
              y=rep(dat$Malform, dat$Freq))
M <- sqrt(sum(df3)-1)*r
p_M <- 2*pnorm(abs(M),lower=F)
```

```
### Table 3 (Ordinal Data)
df3 <- matrix(c(48, 17066, # 0: Yes, No
38, 14464, # <1: Yes, No
5, 788, # 1-2: Yes, No
1, 126, # 3-5: Yes, No
1, 37), # 5<: Yes, No
              ncol = 2, byrow = TRUE,
              dimnames = list(AlcoholUsage = c("0","<1","1-2","3-5","5<"),
                              MalForm = c("Yes","No")))
df3

# Defination of scores
u <- c(0,0.5,1.5,4,7) # AlcoholUsage
v <- c(1,0) # Malform: Yes=1,No=0

# Data Generation
dat <- expand.grid(AlcoholUsage=u, Malform=v)
dat$Freq <- as.vector(df3)
```

R Code (Cont'd)

Ordinal Data

알코올량	기형없음	기형있음	합계	기형비율(%)	수정잔차
0	17066	48	17114	0.28	-0.18
<1	14464	38	14502	0.26	-0.71
1-2	788	5	793	0.63	1.84
3-5	126	1	127	0.79	1.06
5<	37	1	38	2.63	2.71

```
# Measures of association for ordinal data
```

```
require(DescTools)
```

```
GoodmanKruskalGamma(df3)
```

```
#  $\gamma$  (gamma)
```

```
KendallTauB(df3)
```

```
# Kendall's  $\tau_b$ 
```

```
StuartTauC(df3)
```

```
# Stuart's  $\tau_c$ 
```

```
SomersDelta(df3, direction = "row")
```

```
# D(X->Y): Row=Explanatory, Col=Response
```

```
SomersDelta(df3, direction = "col")
```

```
# D(Y->X)
```

R Code (Cont'd)

Three-way Table (CMH)

페니실린	시간	생존여부		합
		치료	죽음	
1/8	즉시	0	6	6
	이후	0	5	5
1/4	즉시	3	3	6
	이후	0	6	6
1/2	즉시	6	0	6
	이후	2	4	6
1	즉시	5	1	6
	이후	6	0	6
4	즉시	2	0	2
	이후	5	0	5

```
# OR according to penicillin
apply(df4+0.5, 3, function(M){
  (M[1,1]*M[2,2])/(M[1,2]*M[2,1])
})
```

```
## Three-way Table
df4 <- array(
  c( 0,0, 6,5, # Penicillin 1/8: [time 0,90min] x [Survival Yes,No]
    3,0, 3,6, # Penicillin 1/4:
    6,2, 0,4, # Penicillin 1/2:
    5,6, 1,0, # Penicillin 1:
    2,5, 0,0), # Penicillin 4:
  dim=c(2,2,5),
  dimnames=list(Time=c("0","90"),
    Survival=c("Yes","No"),
    Penicillin=c("1/8","1/4","1/2","1","4")))
```

df4

```
# CMH test (Conditional Independence)
mantelhaen.test(df4+0.5)
```

```
# Breslow-Day Test (Homogeneity)
bd <- BreslowDayTest(df4+0.5); bd
bd_tarone <- BreslowDayTest(df4+0.5, correct = TRUE); bd_tarone
```